The Green River Affair

A Scientific Mystery in the Kingdom of Zyrkland

Zyrkland

In the isolated Kingdom of **Zyrkland**, science has withered under the rule of its eccentric and authoritarian monarch, **King Dyrt the Unquestionable**. Declaring himself the "Guardian of Truth," he rejects experts, measurements, and anything involving "complicated numbers." Only simple, emotional explanations—broadcast through state loudspeakers—are permitted.

For years, King Dyrt insisted that Zyrkland was "the healthiest country in the world," with rivers that "sparkle with life." He dismisses any reports suggesting otherwise as "fake news" or "foreign tricks".

But then fish began to die along the kingdom's central waterway, once known as the **Green River**, now grimly renamed the **Dead River**. Fish are vital to Zyrkland's culture, cuisine, and economy, and citizens demanded answers.

The King provided one—by inventing a scapegoat.

The Myrtfolk

Zyrkland is home to the **Myrtfolk**, a small and peaceful mountain community known for crafting musical instruments and brewing fermented teas. They live far from the Green River and have no industrial activity, yet King Dyrt seized the opportunity.

In a national address, he proclaimed:

“The fish are not dying from natural events.
No, my good people—the Myrtfolk are poisoning the river with their strange brews and superstitions!”

There was no evidence. But the accusation was repeated endlessly.
Scientists were mocked as “evil number priests,” data was banned from public discourse, and asking for charts became an act of disloyalty.

The Fall of ZEPA

Before Dyrt’s rise, Zyrkland relied on the Zyrkland Environmental Protection Agency **(ZEPA)**, a respected scientific institution monitoring water, air, and climate. However, ZEPA’s transparent reports often contradicted the King’s grandiose claims—and for that, he despised them.

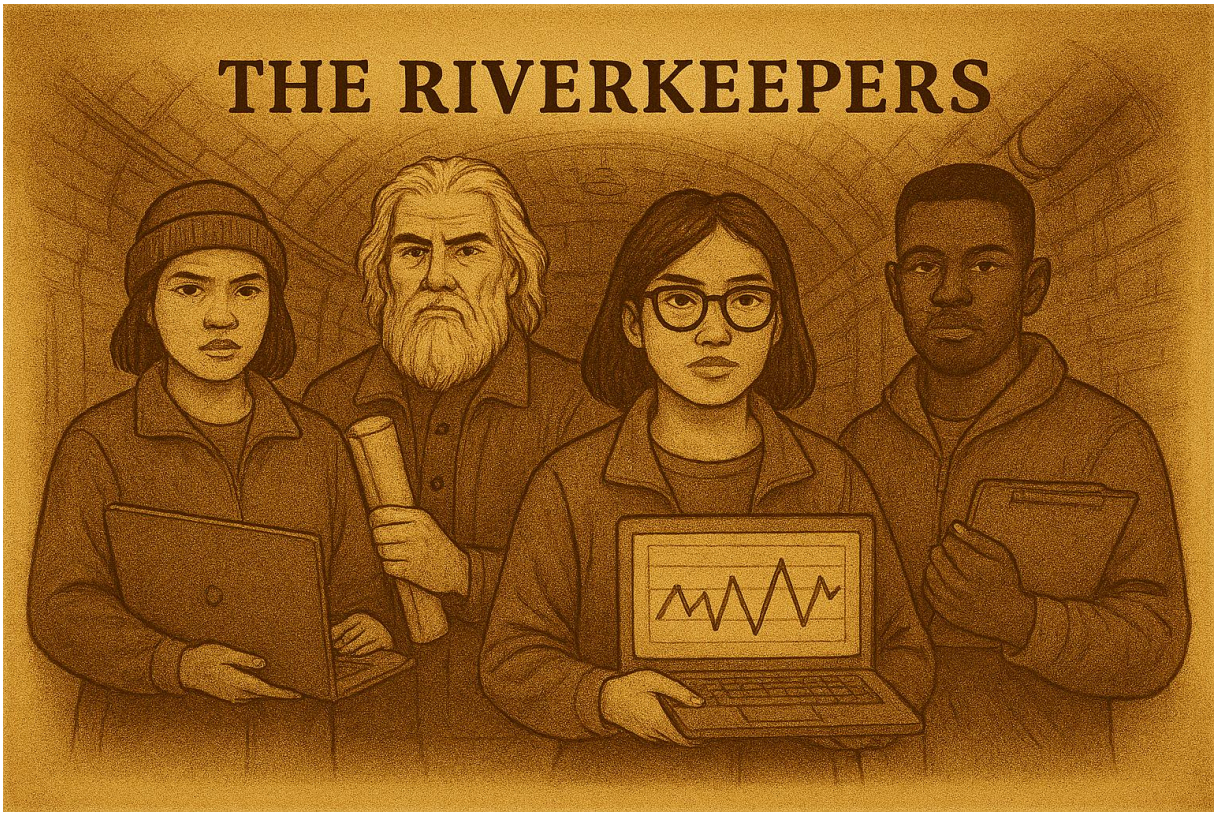
Within a year, Dyrt abolished ZEPA through **Royal Decree Z44-KDUQ**. Some scientists fled the country. Others were arrested by the ZYSPO, Zyrkland’s secret police. Many simply vanished.

But ZEPA’s spirit lived on.

The Rise of the RiverKeepers

Former scientists, IT experts, and engineers regrouped underground, forming the **RiverKeepers**—a clandestine network dedicated to protecting truth. Using forgotten tunnels, abandoned servers, and repurposed sensors, they secretly rebuilt the kingdom’s environmental data infrastructure.

Their motto: *“Truth flows where rivers flow.”*



The Royal Alchemics Combine Factory

Since the King's rise to power, fish mortality along the Green River has intensified. Instead of investigating, Dyrt continued blaming the Myrtfolk—a tidy narrative that strengthened his control.

Meanwhile, the King's most protected industrial asset, the **Royal Alchemics Combine**, operates on the riverbank. Officially, it produces "nutritional supplements for the people of Zyrkland". Unofficially, RiverKeeper intelligence suspects it manufactures **restricted chemical agents** for Zyrkland's army, the Wyrtmacht. These chemical agents could be used in genocidal campaigns against the Myrtfolk and other perceived enemies.

Whether the fish deaths result from natural processes or from the Royal Alchemics Combine's secret activities remains unknown.

That is why **you** have been contacted.

The River Quest

Your team once worked for ZEPA—before its dissolution. When the King outlawed environmental monitoring and arrested your colleagues, you escaped and regrouped underground.

You now serve as part of the **RiverKeepers**, Zyrkland's last line of scientific and technological defense.

The regime destroyed most ZEPA data, but fragments survived. These datasets—carefully smuggled out before ZYSPO agents could erase them—are now in your hands.

Top Secret Mission

Before you proceed, remember this. What you are about to do is highly sensitive, dangerous, and strictly confidential. King Dyrt's regime has erased data, silenced scientists, and destroyed entire archives to keep the truth buried. The RiverKeepers operate in the shadows because discovery means arrest—or worse.

You now hold fragments of knowledge the kingdom was never meant to see. Handle them with caution. Work with precision. And above all, trust the data, not the propaganda.

Your mission is as follows:

- **Explore the surviving meteorological, hydrological, chemical, and biological datasets.**
  Begin by examining what each dataset contains, what the variables mean, and how the different measurements relate to each other. Treat this as your first step in uncovering the river's story.
- **Perform a rigorous Exploratory Data Analysis (EDA) to reveal the river's "data story".**
  Use visualizations and descriptive statistics to uncover patterns, trends, anomalies, and correlations. Be aware that the datasets may contain errors, inconsistencies, gaps, or suspicious values—some caused by failing sensors, others perhaps due to deliberate deletion or negligence. Recognizing and interpreting these issues is part of the investigation.
- **Reconstruct what truly happened during each fish-mortality event.**
  Integrate your findings from all datasets to build an evidence-based explanation—without assuming the cause beforehand. You are piecing together a timeline from incomplete, partially corrupted data, just as skilled data scientists often must.
- **Consider the possibility of unreported or hidden events.**
  The regime destroyed many ZEPA archives and suppressed environmental monitoring. Your analysis should remain open to the idea that some events might not be documented at all, and that the data may hint at incidents the authorities never recorded.
- **Write a clear, structured scientific report that communicates the river's data story.**
  The report should summarize your key findings using clear visualizations and explain how you explored the data, which methods you used, and why. Describe any uncertainties, missing values, or data quality issues you encountered, and how you addressed them. Finally, present conclusions that are directly supported by the evidence you discovered during your analysis.

Your report will be used to alert international scientific and political bodies—and to challenge King Dyrt's authoritarian narrative. You can prove the innocence of the Myrtfolk!

The truth has been buried. But with your skills, it can surface again!

For Zyrkland. For Truth.